

AVG's 'Chime'

Antivirus company is taking help of crowdfunding to fund its project 'Chime' a router. Why? Read here: <http://dgit.in/AVGRouter>

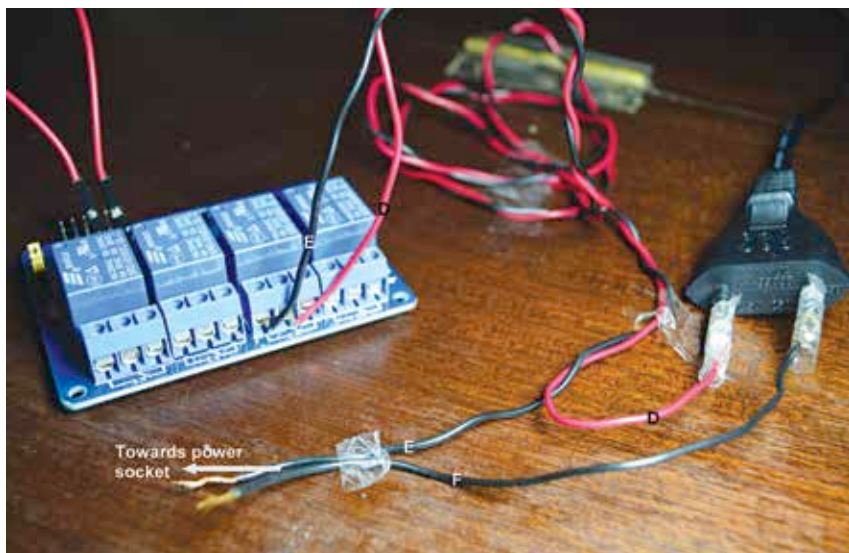


Ford launches Sync Connect

Forget keyless entry, how about starting your car using an app? Exactly what Ford's Sync Connect does <http://dgit.in/FordSyncConct>

So, let's understand the connections. The Digital pin # 8 on the Arduino is connected to the "IN2" pin of the relay (labelled B in Image#3), which is basically used to send commands to the latter. The 'VCC' & the 'GND' pins of the relay are connected to the '5V' and 'GND' ports of the Arduino board respectively and these pins help power up your relay. If you don't have enough 'male-to-female' jumper wires, you could also use a general wire and remove its plastic insulator from both ends – with the help of some duct tape, you'll be good to go! The wires have been marked to help you identify the connections.

Let's connect the appliance to the relay. Before we begin, take a look at the connections in the image below. Take the plug of your appliances and use duct tape to attach two wires, red and black in color (i.e D and F marked wires, respectively). The F wire will



This is how you connect your appliance and the relay to power socket.

directly go in the traditional power socket while the D wire will first go into the relay's 2nd switch's 'Common' port (usually the middle connector of a relay) and a new E wire will be attached onto either the Normally Closed or Normally Open port of the relay, as per your wish. This other end of E wire will then go with the F wire, into the power socket.

Now for the final steps, connect the Arduino with the Computer using a USB and put the E & F wires in the 240 Volt AC power socket. Be extra cautious while handling this part, a slight mistake could even prove fatal.

Now open up your computer and set up the website. If my aim is to control lights from 'anywhere', I'd go ahead with a proper hosting and a domain name. You'd need to create a page named "index.html" that will contain the following code:

You can obviously add the magic of CSS to create something more interactive like <http://www.PraneetSah.com/Arduino/>. Finish up the website by creating the final 'control.php'.

Now upload these two files to your web server, and make sure they are accessible. Open up the website, type http://www.your_website.com/, click on the text: "Click here to turn ON" and then check the directory of these files. You'll notice

that a file called "status.txt" has appeared with its content as a single digit "1" and it would be accessible through http://www.your_website.com/status.txt. You can try again by clicking on the "Click here to turn OFF" text and the 'status.txt' file would be updated with its content as '0'. And with that, we're done setting up our website.

The next step is to program our Arduino through 'Arduino IDE' and write a code in 'Processing IDE' that would check the 'status.txt' file continuously for content change. Open up your Arduino IDE and write the following program in it. I've used the 'blink' example given in the IDE and edited it a little to act as required.

Click on the 'Upload' button in the IDE, and open "serial monitor" from the 'tools' menu. Here, submit capital 'H' or 'L' to check whether the appliance does, in fact, turn on or off, as per the instructions, or not. In most cases, it should, and if it doesn't, check the above steps again. Or if it switches on when "L" is submitted and vice versa, then simply exchange the places of 'H' and 'L' in the program written above. Make sure to close the serial monitor or else it will cause problem with 'processing IDE'.

Finally, open up the "Processing IDE" and type the following code in it. The entire code can be found at this URL: <http://dgit.in/smrthmcode>

In the above code, you would need to modify two things. Firstly, at the place of "Edit_1", you'd need to put a single digit numeric character to which the Arduino is connected. It is basically the "COM port" and you can find it in Arduino IDE → Tools menu → Port. However, if that doesn't work, try the trial and error method, starting from '0' and going up to 15. Another thing you'd need to change is the correct URL

of your 'status.txt' file. Now, simply click on the 'run' button that looks like 'play icon' and you're ready to roll!

Open up your website and your appliances can now be controlled from anywhere! No better way to impress your friends, methinks. On a side note, if you're seriously going to put this into regular practice in your house, then make sure you implement some authorization on the website or else your appliances can be controlled by someone else too. Also, for heavy loads like AC or microwave, our 5V relay won't suffice and will malfunction; therefore, a relay which can handle heavy loads must be used. And lastly, exercise utmost caution in the course of the entire project since you're dealing with extremely high voltage that can even kill you. Good luck, happy building! 

Disclaimer: This project involves working with AC mains power. Unless you have any experience with the same, it is advisable that you only perform the procedures mentioned in the article under supervision of a trained personnel. Improper handling might result in a fatal accident. Digit Magazine and its parent company, Nine Dot Nine Media are in no form responsible for any accidents that may occur during the course of this project or otherwise and is not liable for the same.